FASTLANE ENGINE

Présenté par César **CAPION** & **Lucas ESTRADE**

Quelques *exemples*



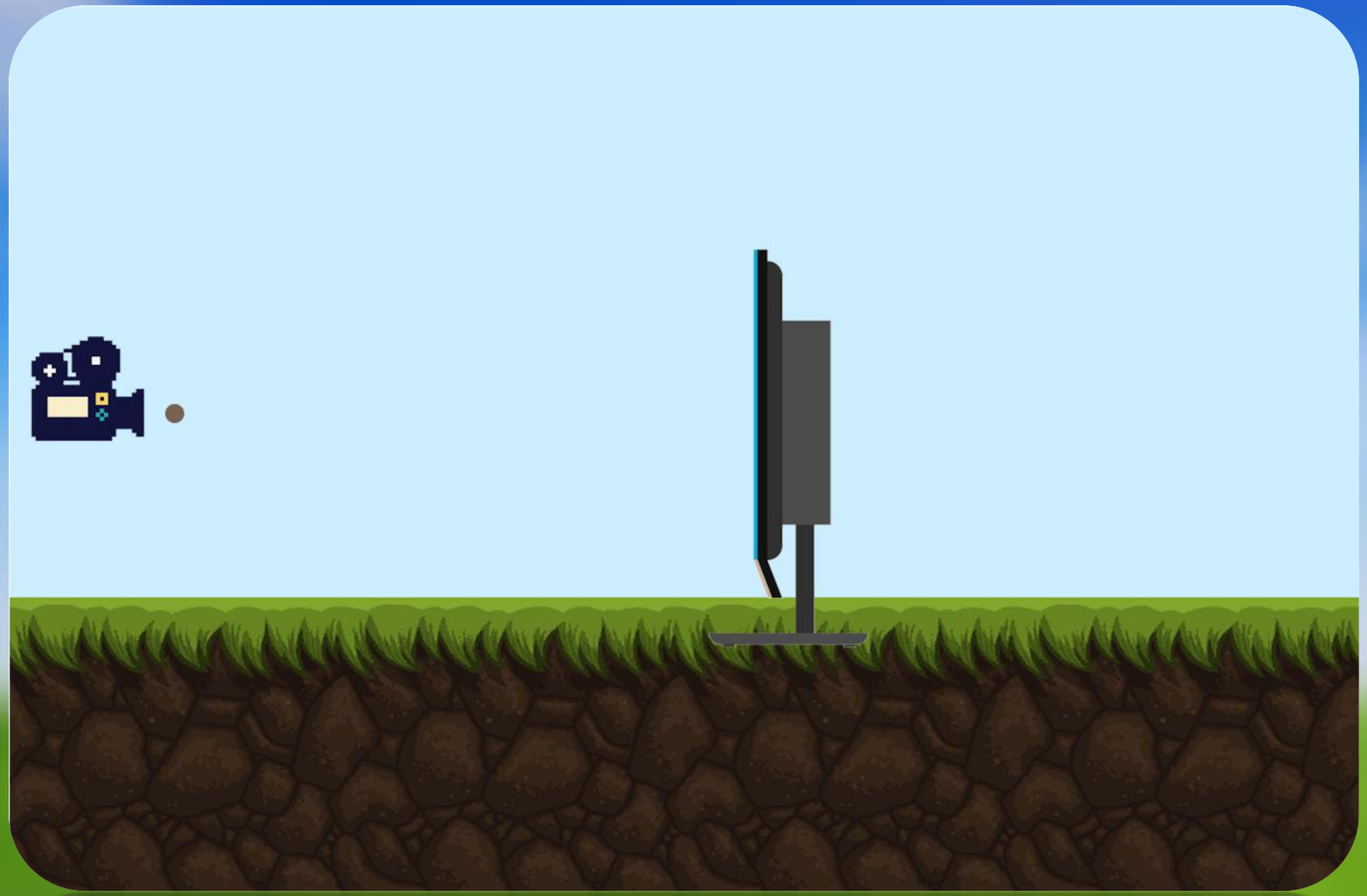
OutRun (1986)



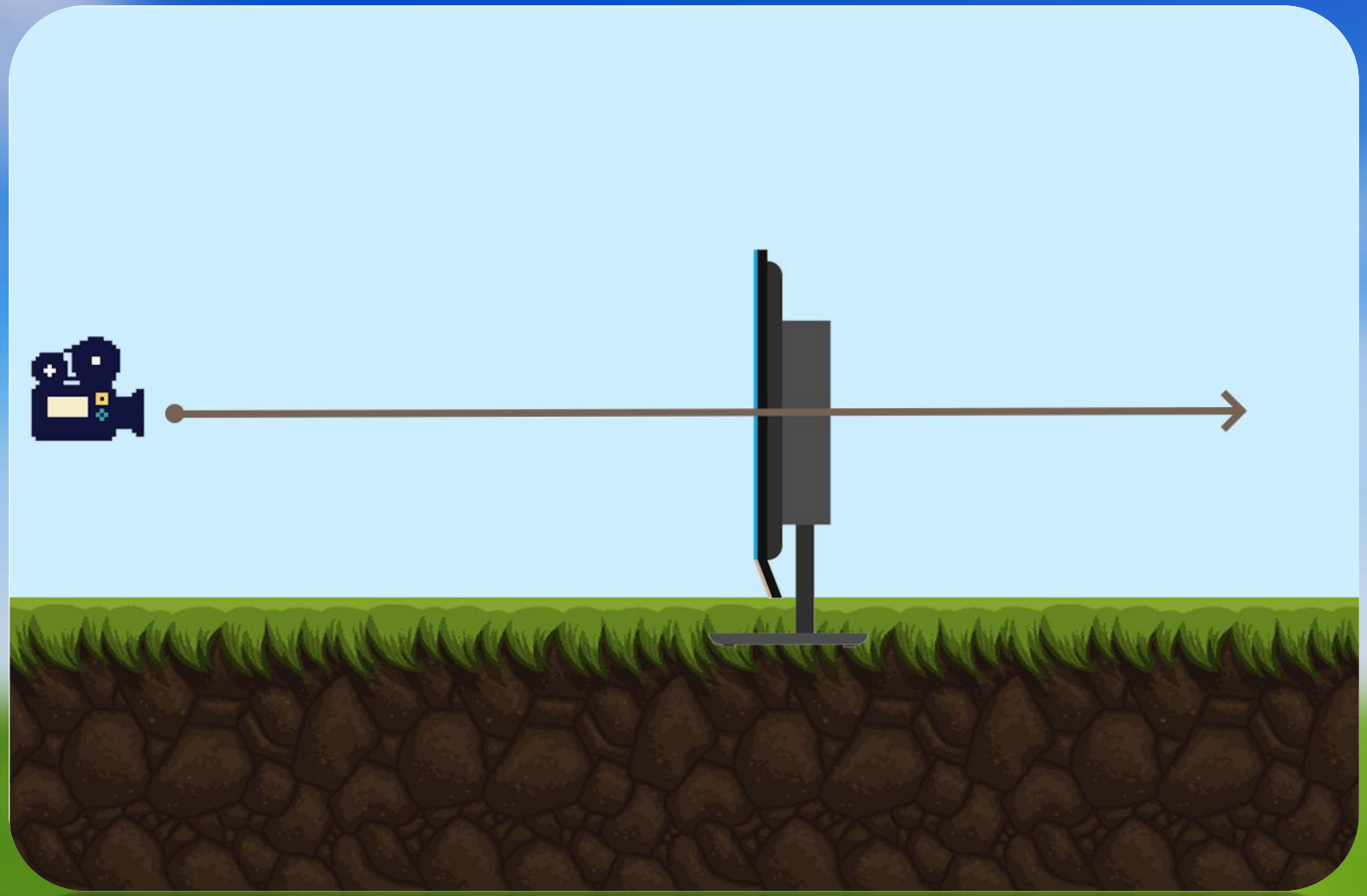
Super Mario Kart (1992)

Concept de base

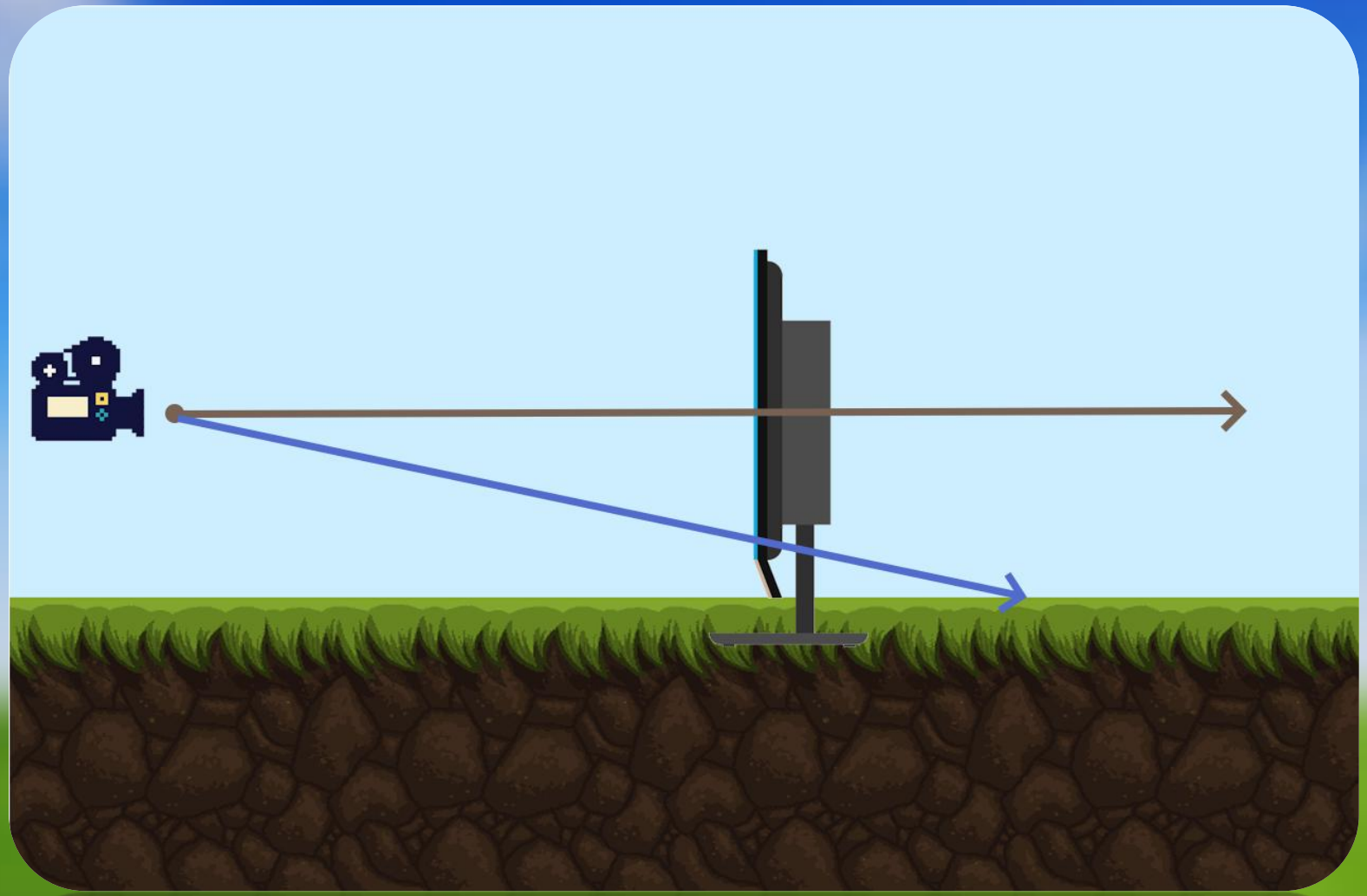
Concept de *base*



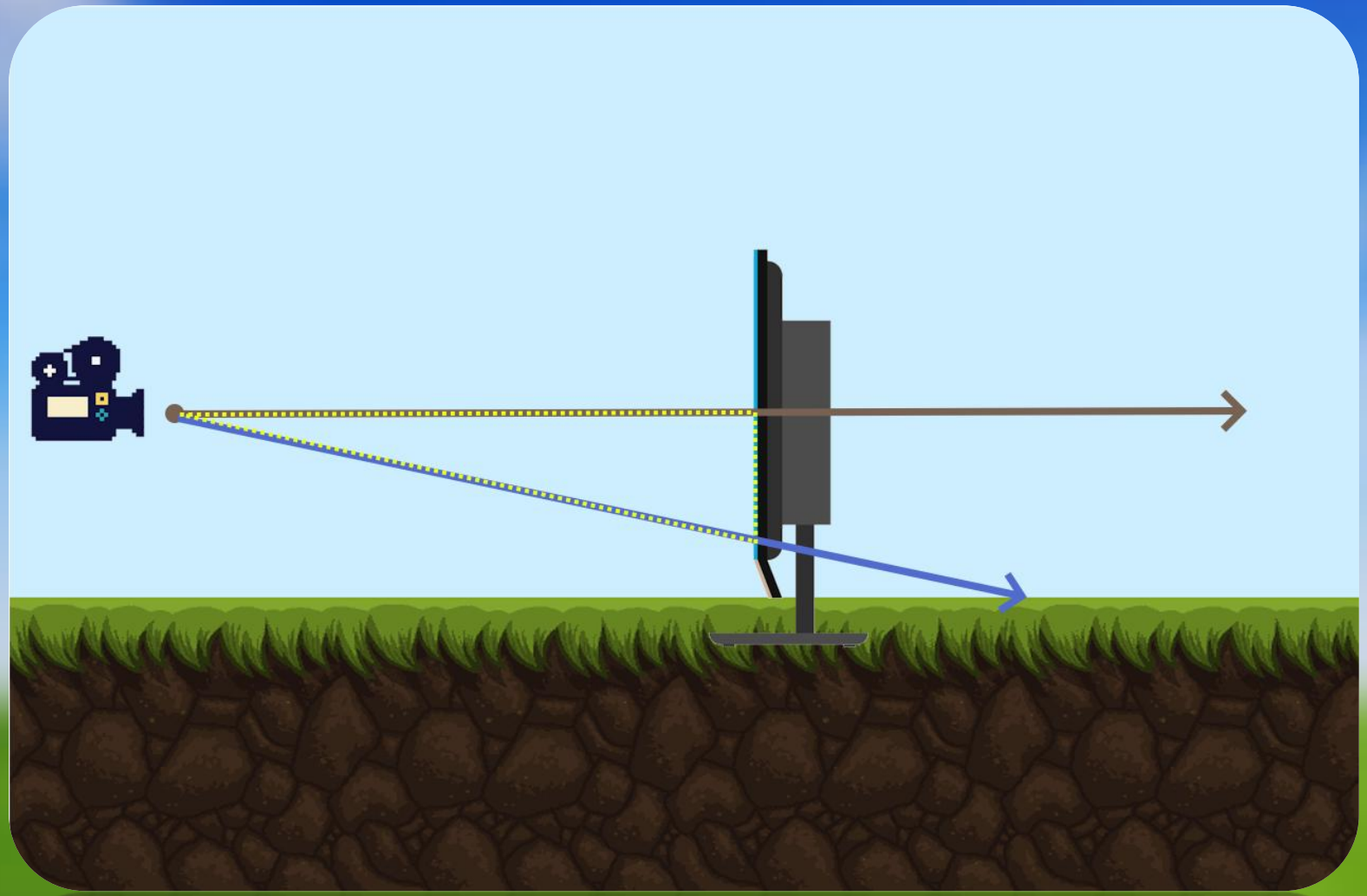
Concept de *base*



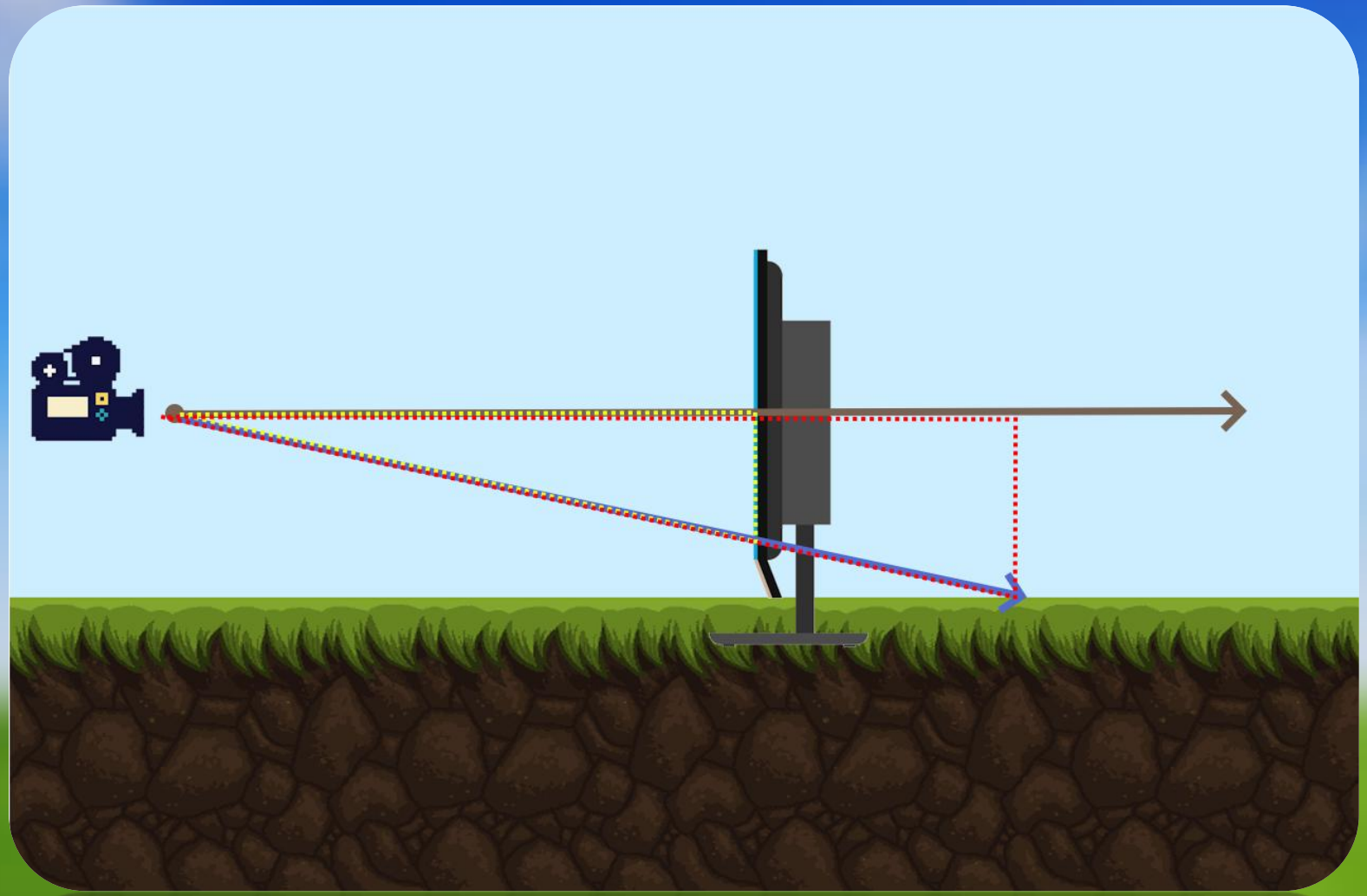
Concept de *base*



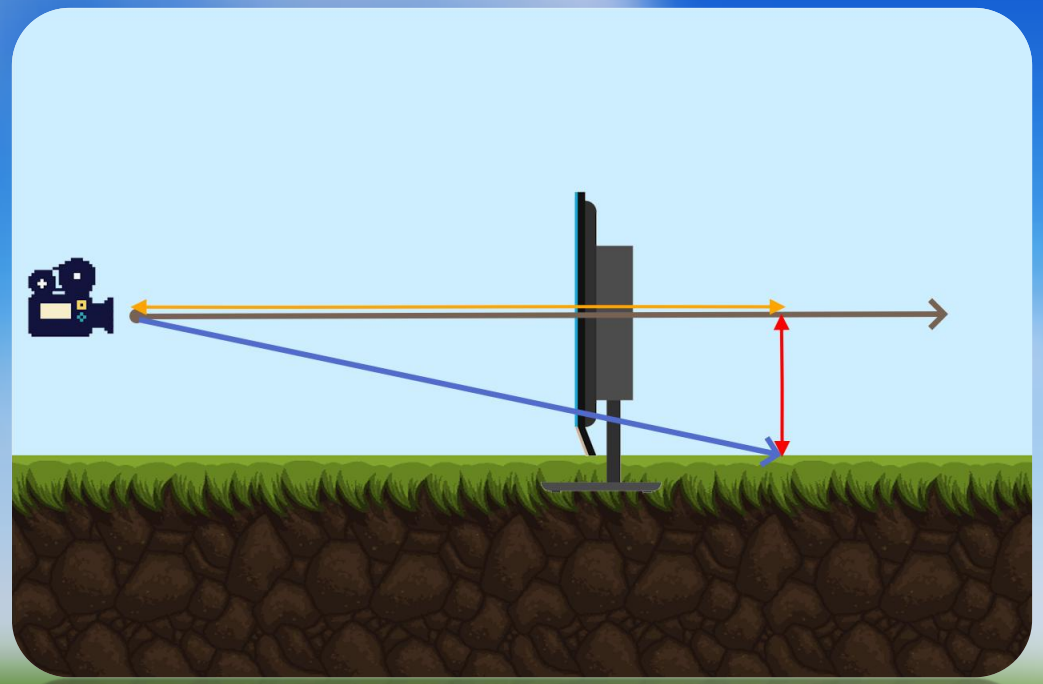
Concept de *base*



Concept de *base*

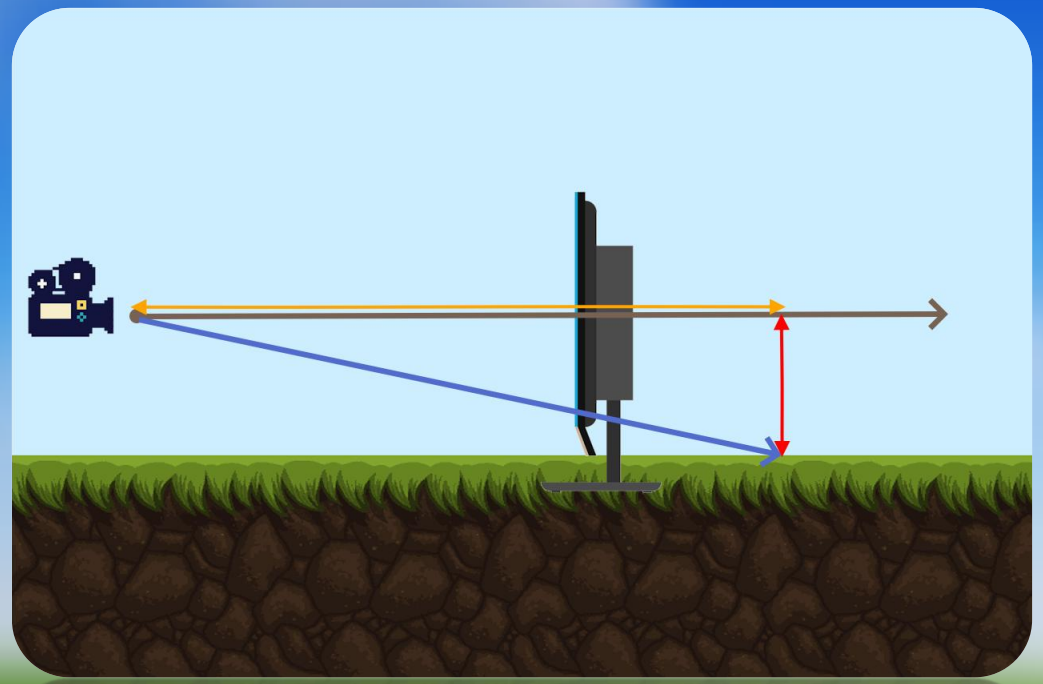


Concept de *base*



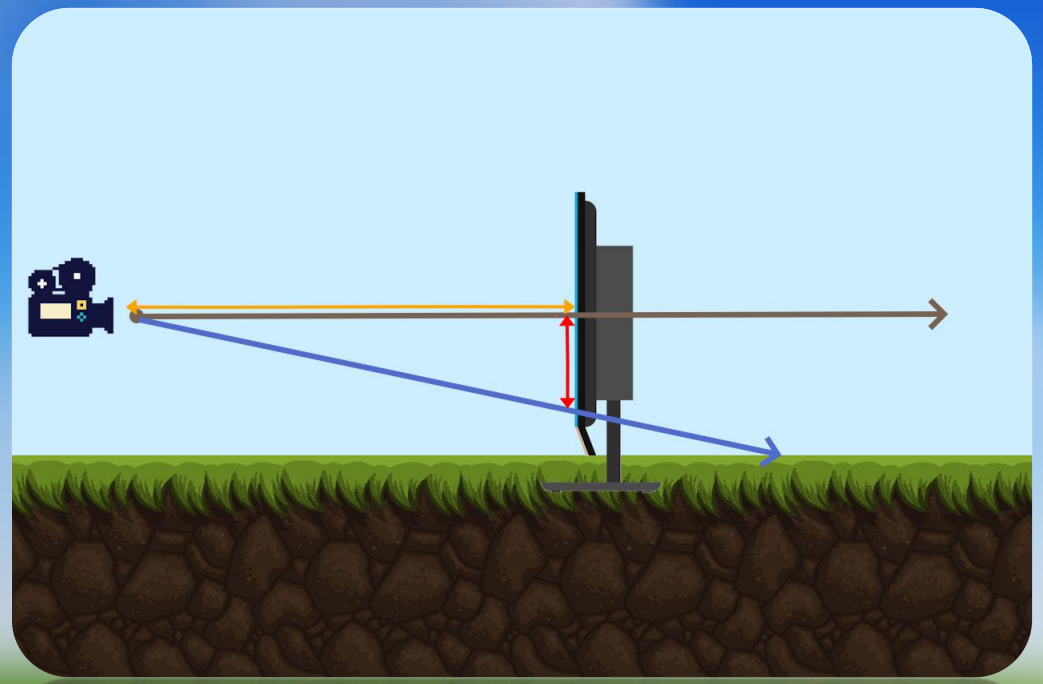
$$\frac{height\ /\ 2}{z}$$

Concept de *base*



$$\frac{cam_height}{z}$$

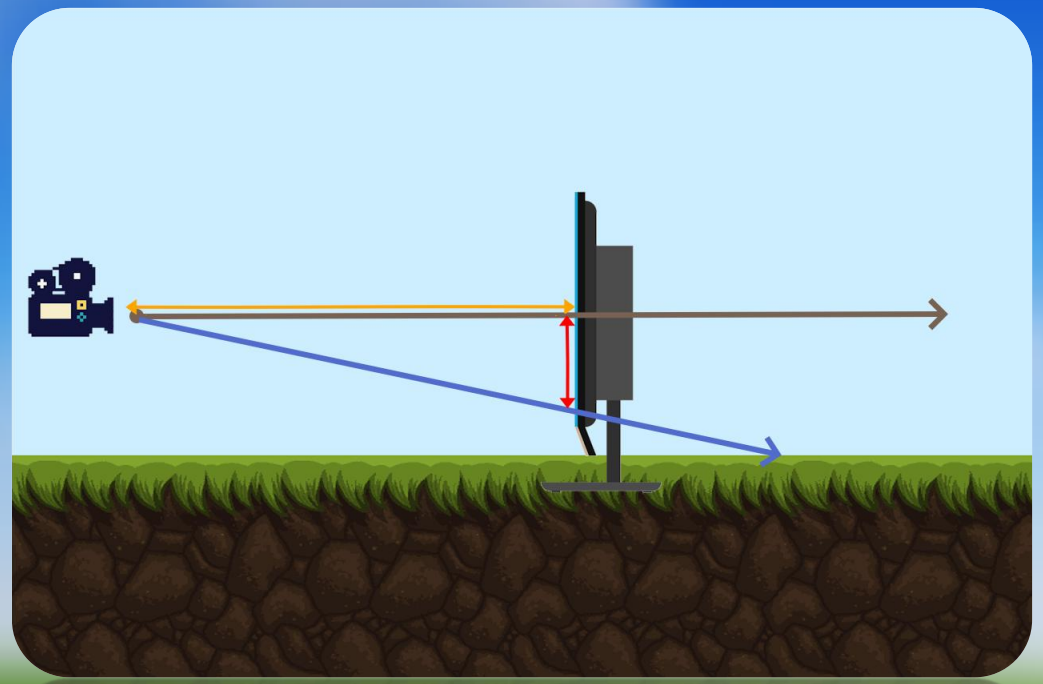
Concept de *base*



$$\frac{cam_height}{z}$$

$$\frac{y_screen - height / 2}{distance}$$

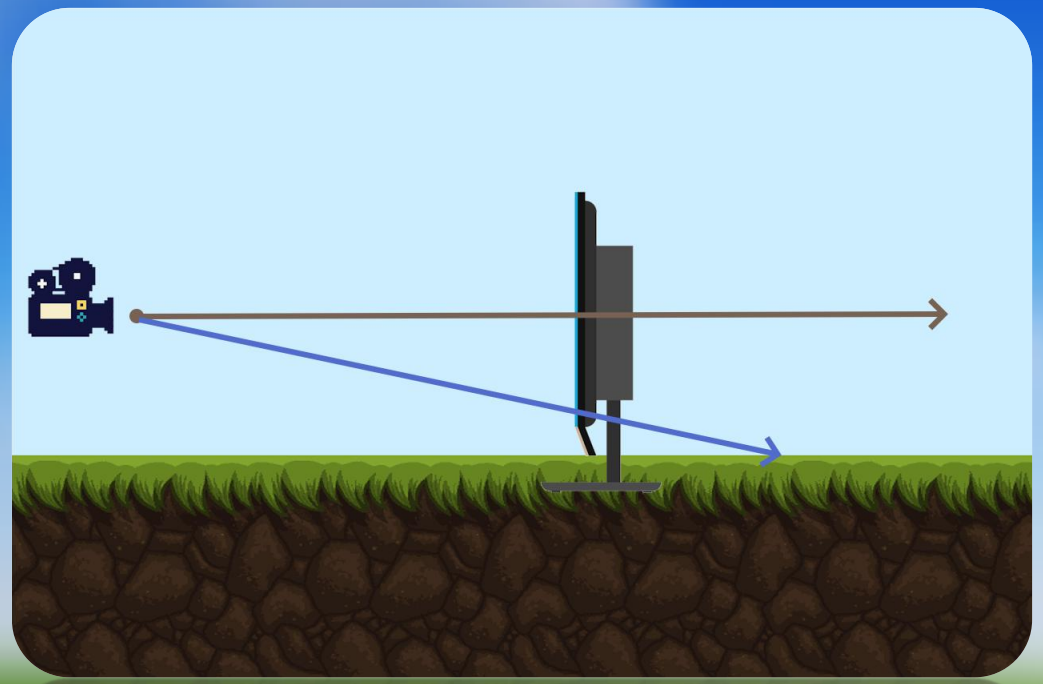
Concept de *base*



$$\frac{cam_height}{z}$$

$$\frac{y_screen - horizon}{distance}$$

Concept de *base*



$$\frac{cam_height}{z}$$

$$\frac{y_screen - horizon}{distance}$$

Concept de *base*

Scanline

$$z = \frac{cam_height \cdot distance}{y_screen - horizon}$$

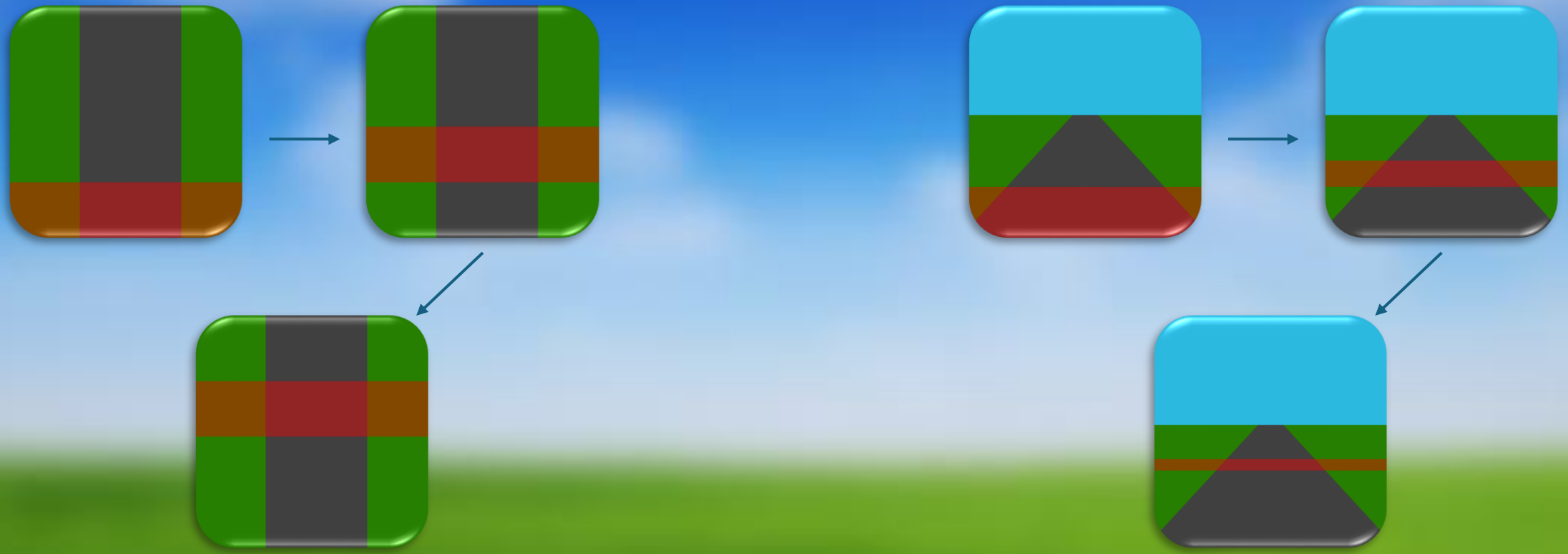
Concept de *base*

Projection

$$y_screen = \frac{cam_height \cdot distance}{z} + horizon$$

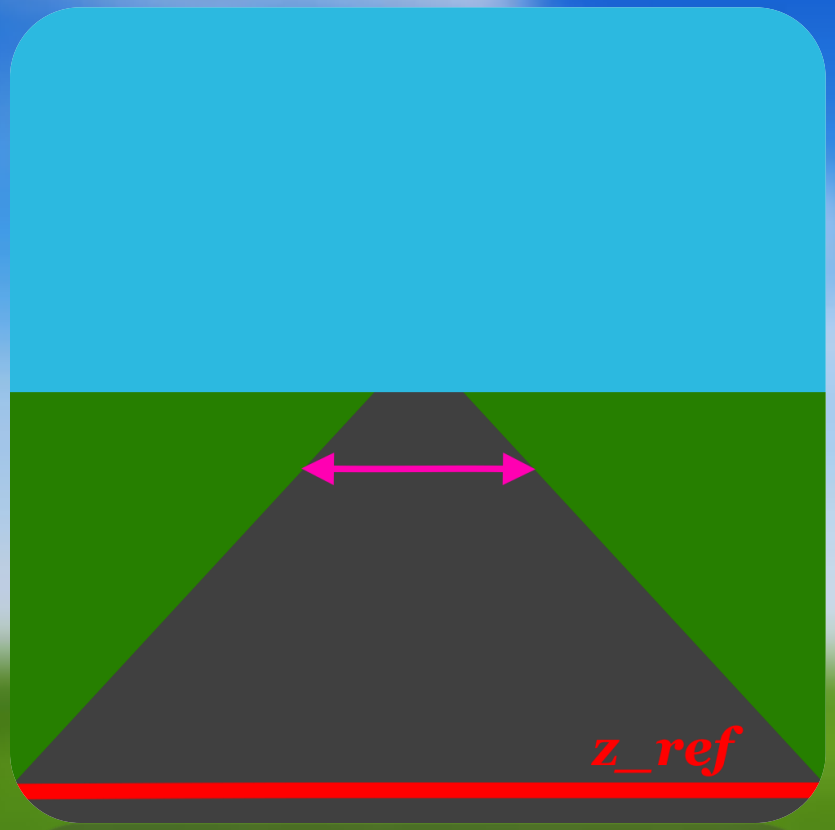
Concept de *base*

Projection



Concept de *base*

Projection



$$road_width_proj = \frac{z_ref \cdot road_width}{z}$$

Virages

Création des routes

Données de piste

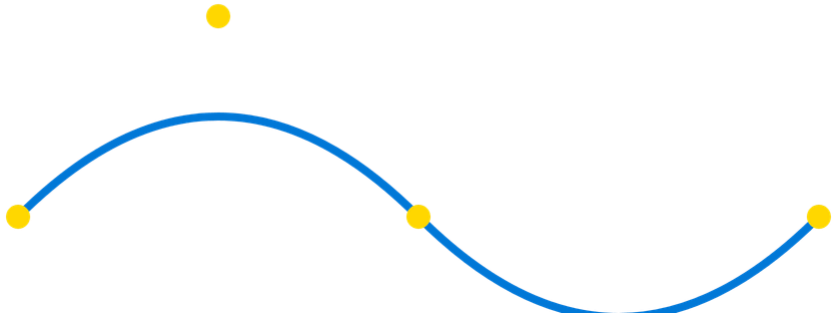
Lecture d'un fichier **txt** au lancement :
- **Altitude** : Variation verticale.
- **Déport** : Position horizontale du centre.

Calcul continu via `trackXAt(z)` avec interpolation précise entre les points de données.

roads > track.txt

```
1   0 0.000000
2   4 0.000000
3   18 0.000000
4   40 0.000000
```

Lissage Catmull-Rom



Évite les "angles" brusques pour une trajectoire fluide et naturelle.

Virages

Géstion de la caméra

POSITIONNEMENT

Gestion de la position latérale du joueur.

Simule le décalage de la voiture sur la largeur de la route pour une conduite active.

SUIVI (LATFOL)

Compensation automatique basée sur **trackXAt(roadPos)**.

Maintient la route centrée à l'écran malgré les virages, assurant un confort visuel optimal.

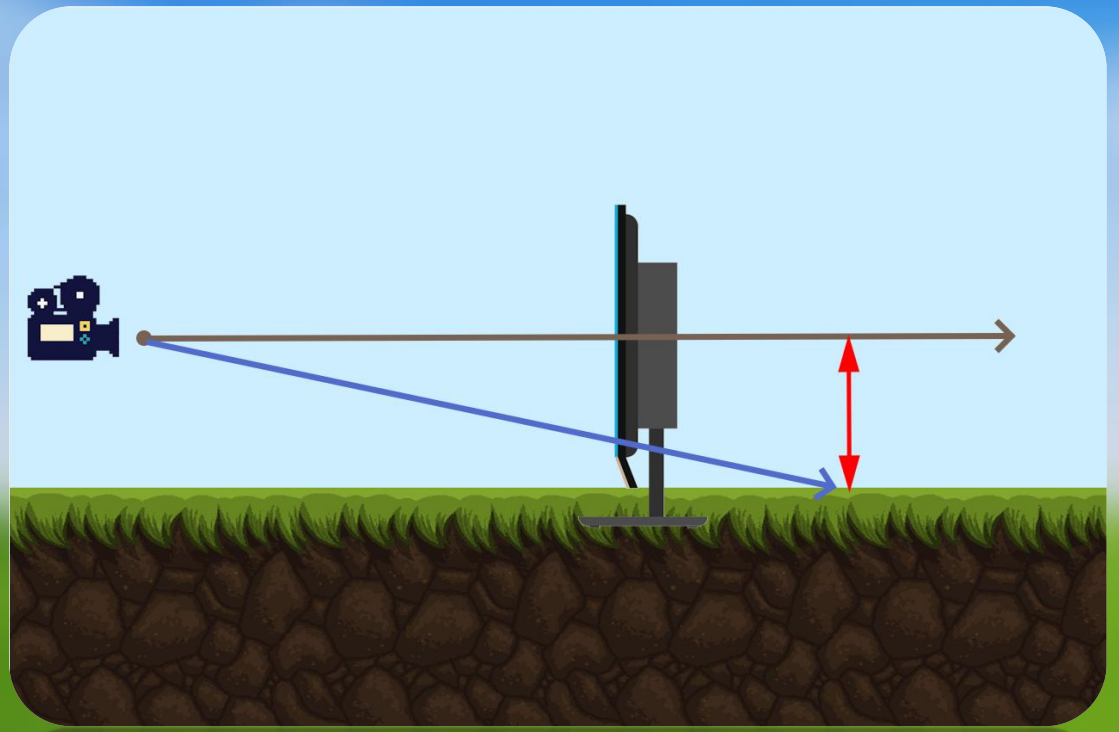
ANTICIPATION

Mode **yawed** utilisant un point distant .

Incline la projection pour "regarder" dans le virage avant d'y entrer, renforçant le réalisme.

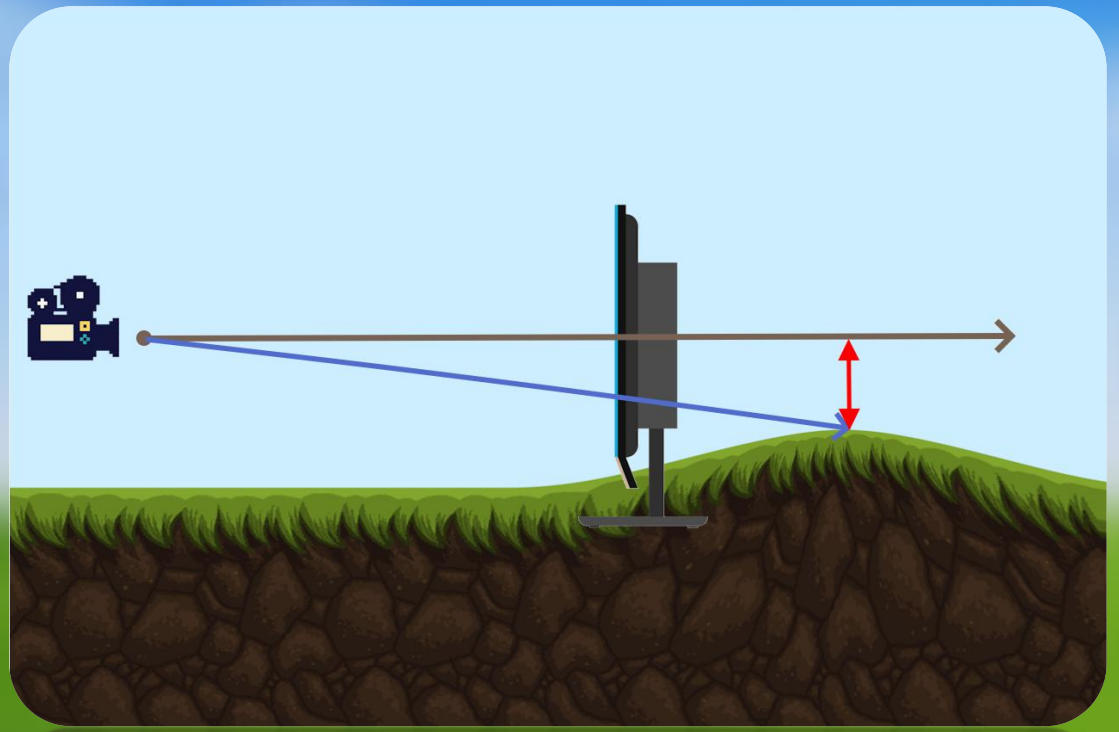
Altitude

$$y_screen = \frac{cam_height \cdot distance}{z} + horizon$$



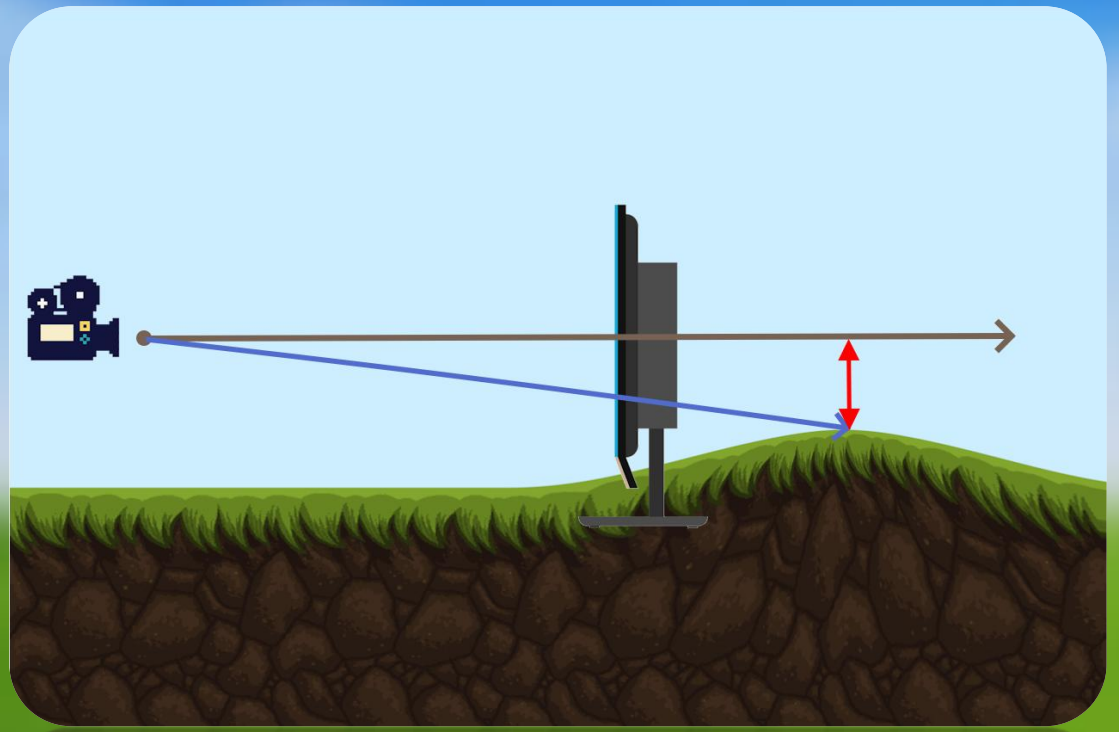
Altitude

$$y_screen = \frac{cam_height \cdot distance}{z} + horizon$$

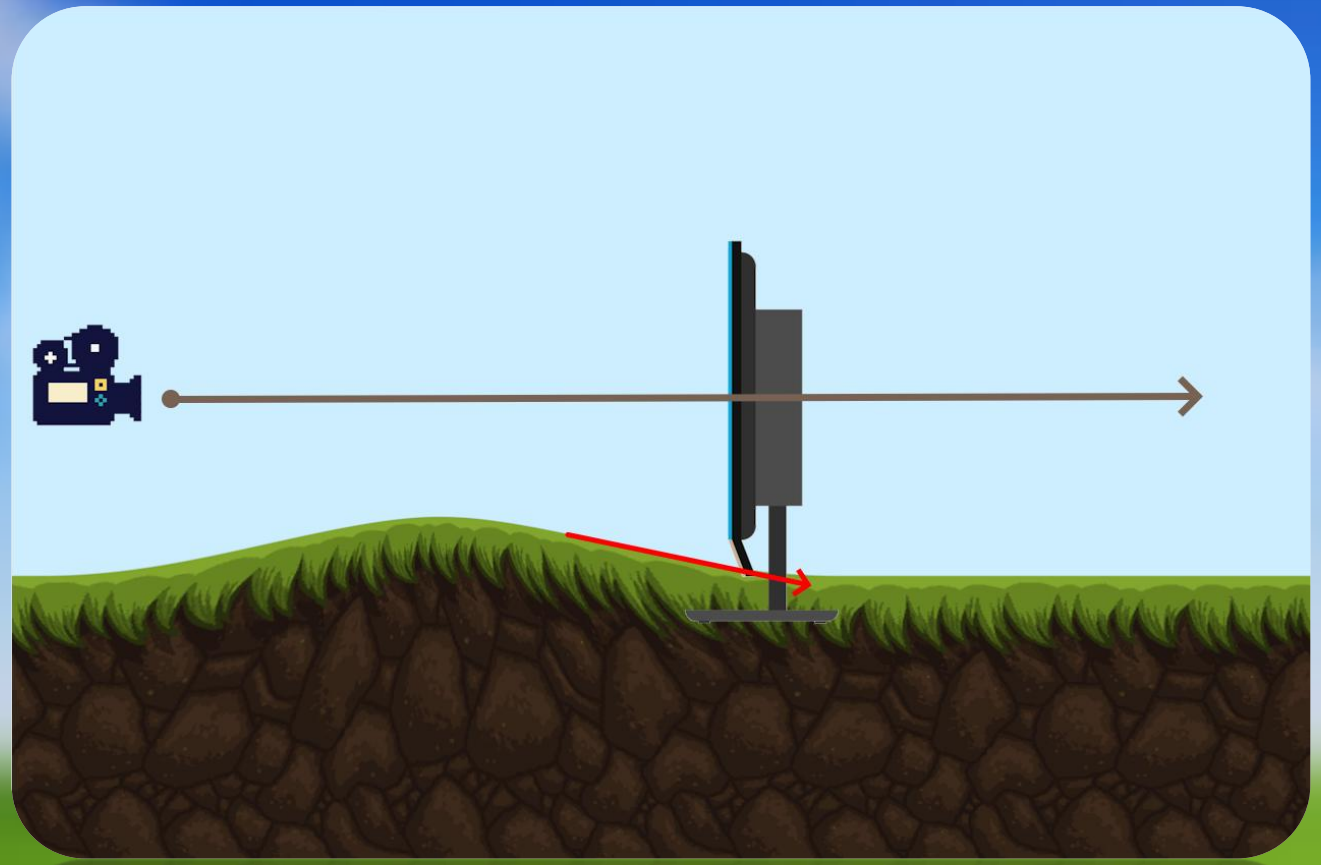


Altitude

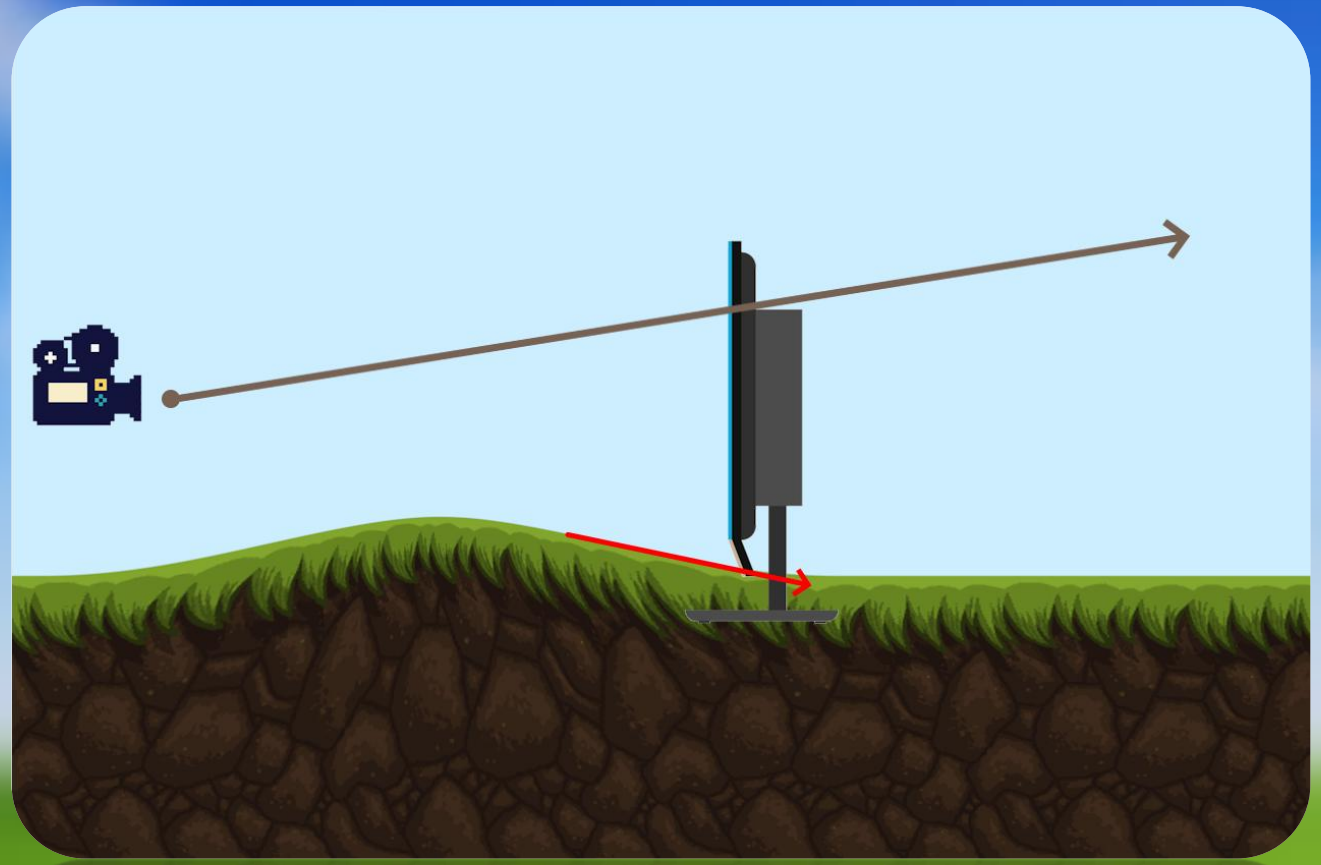
$$y_screen = \frac{\left(cam_height - altitude(z)\right) \cdot distance}{z} + horizon$$



Altitude



Altitude



Apparence *du ciel*

Parallaxe Multicouche

- **SkyTexture** : Nuages, montagnes lointaines et proches.
- **Défilement différentiel** : Les couches proches bougent plus vite que les lointaines.
- **Effet de profondeur** : Création d'une sensation d'espace vaste et dynamique.

Cohérence Visuelle

- **Lien avec les virages** : Décalage horizontal du ciel synchronisé avec la route.
- **Ajustement d'altitude** : L'échantillonnage vertical s'adapte à la position de l'horizon.
- **Immersion totale** : Le décor accompagne naturellement chaque mouvement de caméra.